

CSA 0921 - Programming in Java

Level 2 - Practice Programs

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1.

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name (Vidhushini T.S, 192511061RC). The main section is titled "PRACTICE PROGRAMS" and features a list of questions on the left and a detailed view of the "Q1 Matrix multiplication" program on the right. The program description, code, input, and output are visible.

Q1 Matrix multiplication 10 marks

Matrix multiplication

Your Code

```
1 public class MatMul{
2     public static void main(String[] args){
3         int[][] a={{1,2},{3,4}};
4         int[][] b={{5,6},{7,8}};
5         int r1=a.length,c1=a[0].length,c2=b[0].length;
6         int[][] c=new int[r1][c2];
7         for(int i=0;i<r1;i++){
8             for(int j=0;j<c2;j++){
9                 for(int k=0;k<c1;k++){
10                    c[i][j]+=a[i][k]*b[k][j];
11                }
12                System.out.print(c[i][j]+" ");
13            }
14            System.out.println();
15        }
16    }
17 }
```

Input

stdin input

Output

```
19 22
43 50
```

2.

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name (Vidhushini T.S, 192511061RC). The main section is titled "PRACTICE PROGRAMS" and features a list of questions on the left and a detailed view of the "Q2 . Largest number" program on the right. The program description, code, input, and output are visible.

Q2 . Largest number 10 marks

. Largest number

Your Code

```
1 import java.util.Scanner;
2 public class LargestNumber{
3     public static void main(String[] args){
4         int[] arr={12,23,45,56,78};
5         int max=arr[0];
6         for(int i=1;i<arr.length;i++){
7             if(arr[i]>max){
8                 max=arr[i];
9             }
10        }
11        System.out.println("Largest number: "+max);
12    }
13 }
```

Input

stdin input

Output

```
Largest number: 78
```

Run **Save**

3.

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the title 'SIMATS Online Programming Lab', and the user 'Vidhushini T S' with ID '192511061RC'. The page is titled 'PRACTICE PROGRAMS' and features a 'Back' button. On the left, a 'QUESTIONS' sidebar lists six problems, with 'Q3 Smallest number' selected. The main area displays the question 'Smallest number' (10 marks) and a text input field. Below this is a 'Your Code' section with a Java code editor containing the following code:

```
1 public class SmallNumber{
2     public static void main(String[] args){
3         int[] arr={12,23,45,56,78};
4         int min=arr[0];
5         for (int i=1;i<arr.length;i++){
6             if(arr[i]<min){
7                 min=arr[i];
8             }
9         }
10        System.out.println("Smallest Number: "+min);
11    }
12 }
```

Below the code editor are 'Run' and 'Save' buttons. To the right, an 'Input' section has a 'stdin input' field, and an 'Output' section shows the result: 'Smallest Number: 12'. The bottom of the screen shows a Windows taskbar with a weather widget (31°C, Partly sunny), search bar, and system clock (10:19, 04-08-2026).

4.

The screenshot shows the SIMATS Online Programming Lab interface, similar to the previous one. The 'QUESTIONS' sidebar on the left has 'Q4 . Nth largest number' selected. The main area displays the question '. Nth largest number' (10 marks) with a text input field. The 'Your Code' section contains the following Java code:

```
1 import java.util.Arrays;
2 public class NlargestNumber{
3     public static void main(String[] args){
4         int[] arr={12,23,45,56,4};
5         int n=2;
6         Arrays.sort(arr);
7         int nlargest=arr[arr.length-n];
8         System.out.println(n+"th Largest number is: "+nlargest);
9     }
10 }
```

'Run' and 'Save' buttons are located below the code editor. The 'Input' section has a 'stdin input' field, and the 'Output' section shows: '2th Largest number is: 45'. The bottom of the screen shows the same Windows taskbar with a weather widget (31°C, Partly sunny), search bar, and system clock (10:19, 04-08-2026).

5.

The screenshot shows the SIMATS Online Programming Lab interface. The top navigation bar includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user profile "Vidhushini T S". The main section is titled "PRACTICE PROGRAMS" and displays a list of questions on the left. The selected question is "Q5 Palindrome number/string" with a value of 10 marks. The question description is "Palindrome number/string". The "Your Code" section contains a Java program that checks if a string is a palindrome. The "Input" field is labeled "stdin input" and is empty. The "Output" field shows the result "Palindrome".

QUESTIONS

- Q1 Matrix multiplication 10 marks
- Q2 Largest number 10 marks
- Q3 Smallest number 10 marks
- Q4 Nth largest number 10 marks
- Q5 Palindrome number/string 10 marks
- Q6 Removing duplicates in array 10 marks

6/6 answered

Q5 Palindrome number/string 10 marks

Palindrome number/string

Your Code

```

1 public class Palindrome{
2     public static void main(String[] args){
3         char[] arr={'r','a','c','e','c','a','r'};
4         boolean isPalindrome=true;
5         int left=0;
6         int right=arr.length-1;
7         while(left<right){
8             if(arr[left]!=arr[right]){
9                 isPalindrome=false;
10                break;
11            }
12            left++;
13            right--;
14        }
15        if(isPalindrome){
16            System.out.println("Palindrome");
17        }else{
18            System.out.println("Not a Palindrome");
19        }
20    }
21 }

```

Input

stdin input

Output

Palindrome

6.

The screenshot shows the SIMATS Online Programming Lab interface. The top navigation bar includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user profile "Vidhushini T S". The main section is titled "PRACTICE PROGRAMS" and displays a list of questions on the left. The selected question is "Q6 Removing duplicates in array" with a value of 10 marks. The question description is "Removing duplicates in array". The "Your Code" section contains a Java program that removes duplicates from an array. The "Input" field is labeled "stdin input" and is empty. The "Output" field shows the result "After removing duplicates: 10 20 30 40".

QUESTIONS

- Q1 Matrix multiplication 10 marks
- Q2 Largest number 10 marks
- Q3 Smallest number 10 marks
- Q4 Nth largest number 10 marks
- Q5 Palindrome number/string 10 marks
- Q6 Removing duplicates in array 10 marks

6/6 answered

Q6 Removing duplicates in array 10 marks

Removing duplicates in array

Your Code

```

1 import java.util.Arrays;
2 public class Duplicate{
3     public static void main(String[] args){
4         int[] arr={10,20,30,10,40,20};
5         Arrays.sort(arr);
6         int[] temp=new int[arr.length];
7         int j=0;
8         for(int i=0;i<arr.length-1;i++){
9             if(arr[i]!=arr[i+1]){
10                temp[j++]=arr[i];
11            }
12        }
13        temp[j]=arr[arr.length-1];
14        System.out.print("After removing duplicates: ");
15        for(int i=0;i<j;i++){
16            System.out.print(temp[i]+" ");
17        }
18        System.out.println();
19    }
20 }

```

Input

stdin input

Output

After removing duplicates: 10 20 30 40